Class 4, Week 11: Features and choosing a model

Lecturer guide for CML3112, CML3113 and CML3111. Keep it private.

The class in one paragraph

Which model should we recommend for predicting cooling load, and how do we explain the choice to someone who pays?

Welcome to the last teaching class of the semester. Over three weeks you have built a model, judged it, and learned how to trust it. Today you compare several models and choose one. In real work you will have several candidate models for predicting cooling load, and you must choose one fairly and explain the choice to the client who pays for the cooling plant. The ideas come first, then the lab, then the quiz for this block.

When each cohort meets

Cohort	Programmes	Date	Time and room
CML3112	Civil Engineering and Architecture	Tuesday 24 November 2026	11:00 to 14:00, Room 106
CML3113	Mining and Petroleum Engineering	Friday 27 November 2026	08:00 to 11:00, Room 105
CML3111	Electrical, Mechatronics, Robotics and Communications	Friday 27 November 2026	11:00 to 14:00, Room 115

Before the class

- Open the deck in Slide Show mode and the lab notebook in Colab before anyone arrives. Run the notebook from the top once so you know it works on your account.
- Put the lab notebook and engineering_data.csv in the shared class folder and post the link. Every learner needs both files in Colab, in the Files panel.
- Write the "check your numbers" table on the board before the lab (numbers for your cohort are in the table below).
- Prepare the Block 2 quiz (forms Q9 and Q10) for the end of the class. The deck and the student guide say it opens in the last 20 minutes of the class and each form stays open for 24 hours; the last 20 minutes of the clock are for it. The deck has 32 authored slides, so keep the formula slides brisk.
- The Engineering ML project ran online from Monday 23 November 00:00 to Wednesday 25 November 23:59, Kampala time, and Class 4 content is not in it. For CML3112 (Tuesday 24 November) the window is still open during this class: say so, remind the class of the support hours on the class channel, and do not answer project questions in the room in a way the other cohorts cannot see. Project marks are final by Wednesday 2 December.
- The deck teaches the rule the project states as an instruction (client limits on validation MAE, 2.0 deg C, 2.0 kW or 1.0 kWh/t, and on latency, 50 ms; then the lowest validation MAE; ties by latency, then name). It also recalls the cost table for the threshold (Class 2), shows five-group cross-validation with ddof=1 and the refit on E1 to E5 before the one test opening. The latencies and costs are invented teaching numbers.

The clock

Tuesday cohort (CML3112): the first full teaching of this class. Friday cohorts (CML3113, CML3111): the same deck and the same clock, because from Class 2 on all three cohorts are on the same footing.

CML3112, Tuesday

Starts	Minutes	What happens
0:00	10	Recall from Class 3
0:10	35	Concepts, part 1: features, scaling and two new models

Starts	Minutes	What happens
0:45	35	Concepts, part 2: choosing a model, the test set, the memo
1:20	15	Break
1:35	55	Lab: the comparison table and the client's rule
2:30	10	Close and questions
2:40	20	Block 2 quiz: the forms open (they stay open 24 hours)

CML3113 and CML3111, Friday

Starts	Minutes	What happens
0:00	10	Recall from Class 3
0:10	35	Concepts, part 1: features, scaling and two new models
0:45	35	Concepts, part 2: choosing a model, the test set, the memo
1:20	15	Break
1:35	55	Lab: the comparison table and the client's rule
2:30	10	Close and questions
2:40	20	Block 2 quiz: the forms open (they stay open 24 hours)

Slide map

The deck has two halves. Slides up to the divider are concepts: no laptops. After the divider come the lab slides and the notebook. You switch once. The titles below are from the CML3112 deck. The other two decks have the same slides in the same order; where a title carries a number, their own number replaces it (see the per-cohort table).

Slide	Title	Part	Friday recap
1	Features and choosing a model	Concepts	
2	Today in one view: what you will leave with	Concepts	
3	Your three hours: models and memo first, then the lab	Concepts	
4	Three questions from last time	Concepts	
5	Words we use today	Concepts	
6	Class 3 in short: overfitting, cross-validation and data leakage	Concepts	
7	In real work you compare several models, choose one and explain why	Concepts	
8	A new feature can carry what you know about the building	Concepts	
9	The new feature is a product, and the extended model gets one more weight	Concepts	
10	A feature ablation tests one change on its own, with everything else kept the same	Concepts	
11	Removing ambient_c makes the error clearly worse, and the new features do not help	Concepts	
12	One-hot encoding turns a categorical feature into 0 and 1 columns, and equipment_id stays out of the features	Concepts	
13	Standardisation puts every feature on the same ruler	Concepts	
14	Ridge regression adds an L2 penalty on large weights, which can stop a flexible curve fitting noise	Concepts	

Slide	Title	Part	Friday recap
15	Ridge regression minimises squared errors plus alpha times squared weights	Concepts	
16	A bigger alpha shrinks the weights, and here the validation error grows	Concepts	
17	k-nearest neighbours (k-NN) predicts from the k training rows most like the new row	Concepts	
18	k-NN measures distance on the standardised features, then averages the k nearest targets	Concepts	
19	The size of k matters: too small chases noise, too large flattens the pattern	Concepts	
20	A fair model comparison keeps four things the same for every model	Concepts	
21	Every model also gets a four-fold cross-validated mean and a sample standard deviation	Concepts	
22	Model comparison table: five candidate models on the same 15 validation rows	Concepts	
23	Latency is the time a model takes to give one prediction, and the client sets a limit on it	Concepts	
24	Model selection follows the client's rule: meet both limits, then take the lowest validation MAE	Concepts	
25	Linear regression meets both limits and has the lowest validation MAE, so it is chosen	Concepts	
26	The warning threshold is chosen by cost: 0.3 has the lowest cost on the validation rows	Concepts	
27	The comparison has four limits, and you must say them out loud	Concepts	
28	After the choice, group cross-validation can hold out each of E1 to E5 in turn	Concepts	
29	The test set is opened once: refit the chosen model on E1 to E5, then score E6	Concepts	
30	A decision memo fits on one page, has seven parts and recommends a model and a threshold	Concepts	
31	Read the recipe before you write it: the client's rule in four lines of Python	Concepts	
32	Today in five lines: test ideas, compare fairly, choose by the rule, test once, write it down	Concepts	
33	Part 2: the lab	Lab	
34	The lab has twelve notebook steps in five groups, and the client's rule is the centre	Lab	
35	Check your numbers against these before you move on	Lab	
36	Four errors you will meet, what they mean and what to do	Lab	
37	How this class counts towards your coursework	Lab	
38	Save the notebook and keep your memo	Lab	

Questions to ask, and a good answer

Ask	A good answer
What is a feature ablation?	Change one thing, keep everything else the same, and compare on the same validation rows.
What is an interaction feature?	A column made by multiplying two features, here load_pct times ambient_c.

Ask	A good answer
Why do we leave equipment_id out of the features?	The model must work on new equipment. A group it never saw would have no column of its own.
Why do ridge regression and k-NN need standardised features?	The penalty and the distance both depend on units, so a feature with big numbers would dominate.
What does a bigger alpha do?	Alpha is the regularisation strength. A bigger alpha makes the L2 penalty stronger and the curve flatter. Too big and the model underfits.
How does k-NN measure closeness?	By the Euclidean distance on the standardised features: square the gaps, add them, take the square root.
What is latency, and why is k-NN slow?	The time a trained model takes to give one prediction. k-NN measures the distance to every training row for each new row.
How does the project choose a model?	Keep the models with validation MAE at most the client's limit and latency at most 50 ms, then take the lowest validation MAE; a tie goes to the faster model, then the name.
Why do we divide by k minus 1 for the standard deviation of the folds?	It is the sample standard deviation, np.std(scores, ddof=1), which the project asks for. StandardScaler, in contrast, divides by n.
How is the warning threshold chosen?	Cost = 5 x FN + 1 x FP on the validation rows for 0.3, 0.5 and 0.7; the lowest cost wins, and a tie goes to the higher threshold.
Who may open the test set, and when?	Nobody until the model and the threshold are chosen, and then only once, after refitting on E1 to E5.

Ideas learners often get wrong

What a learner may say	What to say back
A clever new feature must help	Test it. Here the load_x_ambient column changes the error by hardly anything.
The model with the lowest number on the table is the one to choose	Only if it meets the client's limits on error and latency. For CML3113, k-NN has the lowest validation MAE but takes 60 ms, so linear regression is chosen.
Equipment_id should be a feature because it is in the file	The model must work on equipment it has never met, and a new group has no column of its own.
A bigger alpha or a bigger k is always safer	Too much penalty or too many neighbours flattens the pattern. The error goes up again.
The test set can be used to pick between two close models	It is opened once, after the model and the threshold are chosen. After that it cannot give a fair number.
Refitting on E1 to E5 before the test is cheating	The choice is already fixed, so E5 cannot bias it. More rows make a better model, and E6 is still unseen.

Walking the lab

Step	What to look for
Step 1: prepare	Rows left to model with: 88. Rows 58, 15, 15. Anyone with different numbers has run a cell twice or changed a cleaning rule.
Step 2: first loop	Three models, one loop (pipeline with median imputation). Baseline model, linear regression and decision tree match the Class 1 model comparison table. Ask why the loop keeps the comparison fair.

Step	What to look for
Step 3: features	Four versions of the linear regression model. Removing ambient_c makes the error clearly worse. The product and the squared load change the MAE by less than 0.03. Resist the urge to explain the product away: the lesson is that the data decide.
Step 4: one-hot encoding	The validation group has an is_E5 column that the training table lacks. This is the whole argument for leaving equipment_id out.
Step 5: standardisation and ridge regression	Ridge regression on two features gets slightly worse as alpha grows. The degree-6 curve on load alone improves at alpha 1 (CML3113: no better) and worsens at alpha 100, and its biggest weight shrinks from thousands to single figures.
Step 6: k-NN	k = 3 is best of 1, 3 and 20. The three neighbours of the first validation row are printed, and their average is the prediction.
Step 7: the model comparison table	Five rows, with the cross-validated mean and the sample standard deviation (ddof=1). The GroupKFold fold order is chosen by scikit-learn; only the set of groups matters.
Step 8: the client's rule	The candidates are records (a list of dictionaries) built from the Step 7 results. A list comprehension keeps the models within both limits, and sorted with key=lambda ranks them. Class 3 practised the records form. Linear regression is chosen in every cohort; for CML3113 k-NN has the lowest MAE but fails the 50 ms limit.
Step 9: the threshold by cost	The cost table of Class 2 for 0.3, 0.5 and 0.7. CML3111 and CML3112 choose 0.3, CML3113 chooses 0.5.
Step 10: five-group CV and the test set	Five folds E1 to E5 on the rows with no missing value, sample standard deviation. Then one refit on E1 to E5 and one test opening; the test numbers are in the table below for you, and are not on the slides.
Steps 11 and 12: memo and save	Collect one memo sentence from three learners. Insist on restart and run all before saving.

The numbers each cohort should see

Check	CML3112 (kW)	CML3113 (kWh/t)	CML3111 (deg C)
Training, validation, test rows	58, 15, 15	58, 15, 15	58, 15, 15
Linear regression, two features, MAE	0.74 kW	0.67 kWh/t	0.78 deg C
Linear regression plus load_x_ambient, MAE	0.75 kW	0.67 kWh/t	0.80 deg C
Linear regression plus load_squared, MAE	0.75 kW	0.67 kWh/t	0.78 deg C
Linear regression with load_pct only, MAE	3.12 kW	0.98 kWh/t	4.30 deg C
Ridge regression, degree 6 on load alone, alpha 1, MAE	3.05 kW	0.99 kWh/t	4.20 deg C
k-NN, k = 3, MAE	1.16 kW	0.63 kWh/t	1.58 deg C
Ridge regression, two features, alpha 1, MAE	0.84 kW	0.70 kWh/t	0.93 deg C
Decision tree depth 4, MAE (Class 3 tuned depth and its MAE)	3.39 kW (depth 7: 2.62)	1.20 kWh/t (depth 5: 1.12)	4.79 deg C (depth 7: 3.91)
Extended linear regression, intercept b_0 and weights b_1, b_2, b_3	4.8959, 0.3912, 0.6196, 0.00221	1.9344, 0.1074, 0.1276, 0.000506	6.3765, 0.6181, 0.8657, 0.00307

Check	CML3112 (kW)	CML3113 (kWh/t)	CML3111 (deg C)
Ridge regression loss at alpha 1 (squared errors + L2 penalty)	50.4 + 110.6 = 161.1	35.7 + 7.7 = 43.4	65.3 + 262.5 = 327.8
k-NN worked example, nearest distance and prediction	d = 0.129, prediction 47.5 kW	d = 0.129, prediction 13.1 kWh/t	d = 0.129, prediction 70.0 deg C
Linear regression, 4-fold CV MAE: mean, sample standard deviation (lowest to highest)	0.76, 0.01 (0.74 to 0.77)	0.70, 0.09 (0.57 to 0.78)	0.79, 0.06 (0.75 to 0.87)
Lowest validation MAE among the four models	Linear regression	k-NN, k = 3	Linear regression
Client limit on MAE; models kept by the rule; chosen	2.0 kW; Linear regression, Ridge regression, alpha 1; Linear regression	1.0 kWh/t; Linear regression, Ridge regression, alpha 1; Linear regression	2.0 deg C; Linear regression, Ridge regression, alpha 1; Linear regression
Threshold chosen by cost (cost)	0.3 (1)	0.5 (1)	0.3 (0)
Five-group CV (E1 to E5), linear regression: mean and sample standard deviation	0.68 and 0.10 kW	0.68 and 0.10 kWh/t	0.68 and 0.10 deg C
Lab step 10 only, never on a slide: test MAE after the refit on E1 to E5; warning recall and precision	0.47 kW; 1.00 and 0.75	0.48 kWh/t; 0.75 and 1.00	0.46 deg C; 1.00 and 0.75

If things go wrong

- If the concepts run over, the ridge regression and k-NN formula slides are previews: read the formula aloud, point at one worked line and move on. Keep the slides on latency, the client's rule, the cost table, the five-group cross-validation and the test set: they pull Classes 2 and 3 together into one model choice.
- If only forty minutes of lab remain, do steps 1, 2, 7, 8, 9 and 10, and set the feature ablation, ridge regression and k-NN as practice at home.
- If wifi fails in the room, the notebook runs unchanged in VS Code, as in Week 2.
- If several learners are stuck at the same cell, stop the room, run that cell on the projector and let them copy the fix.
- If time runs short before the quiz, cut the memo write-up and ask learners to finish it at home.

After the class

- Share the lab notebook and engineering_data.csv again so absent learners can follow.
- Remind the class when project marks will be final (by Wednesday 2 December).
- Note any question you could not answer and answer it on the class channel.

Speaker script

Every slide carries its spoken script in the PowerPoint notes (open Presenter View). The scripts are first person and are meant to be read or paraphrased. Pauses are marked by questions.